

TUGAS 3

No. _____

Date _____

Tugas 3

1. $f(x) = \sqrt{2x+4} - 3$

a) Domain fungsi $D(f)$

$$f(x) = \sqrt{2x+4} - 3$$

$$2x+4 \geq 0$$

$$2x \geq -4$$

$$x \geq -2$$

jadi Domain $D(f) = \{x | x \geq -2, x \in \mathbb{R}\}$

b) Range $R(f)$

$$R(f) = \{y | y \geq 0, y \in \mathbb{R}\}$$

2. $y = \begin{cases} -x, & x < 0 \\ 3, & x \geq 0 \end{cases}$

a) Tabel

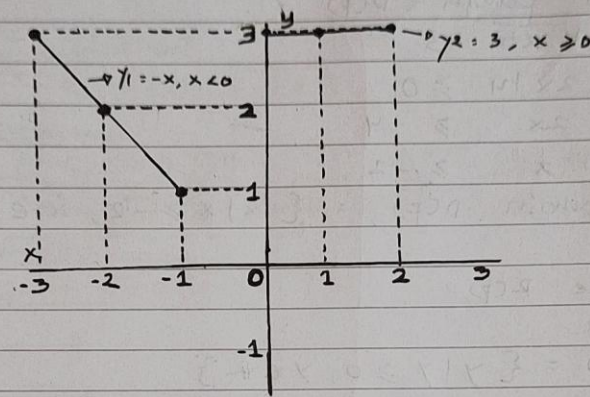
$\Rightarrow y_1 = -x, x < 0 \quad D(f) = \{x | x < 0\} = (-\infty, 0)$

x	y	x,y
-1	1	(-1,1)
-2	2	(-2,2)
-3	3	(-3,3)

$\Rightarrow y_2 = 3, x \geq 0 \quad D(f) = \{x | x \geq 0\} = [0, \infty)$

x	y	x,y
0	3	(0,3)
1	3	(1,3)
2	3	(2,3)

b) Gambar Grafik :



3. $F(x) = 3x^2 + 1$
 $g(x) = \sqrt{3x+6}$

a) $h(x) = f \circ g(x)$
 $= f(g(x))$
 $= f(\sqrt{3x+6})$
 $= 3 \cdot (\sqrt{3x+6})^2 + 1$
 $= 3 \cdot (3x+6) + 1$
 $= 9x + 18 + 1$
 $= 9x + 19$

b) Domain $D(h(x))$
 $\Rightarrow f(x) = 3x^2 + 1$
 $D(f) = \{x | x \in \mathbb{R}\}$
 $\Rightarrow g(x) = \sqrt{3x+6}$
 $D(g) = 3x+6 \geq 0$
 $D(g) = 3x \geq -6$
 $D(g) = x \geq -2$
 $D(g) = \{x | x \geq -2\}$
 $= [-2, \infty)$

$\Rightarrow D(f \circ g) = \{x | x \in \mathbb{R}\} \cap [-2, \infty)$
 $= [-2, \infty)$

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$$4. A = \left\{ \frac{3}{n} \mid n \in \mathbb{N} \right\}$$

$$B = \{n^2 \mid n \in \mathbb{N}\}$$

$$\Leftrightarrow \text{Anggota himpunan } A = \left\{ 3, \frac{3}{2}, 1, \frac{3}{4}, \frac{3}{5}, \dots \right\}$$

$$\text{Anggota himpunan } B = \{1, 4, 9, 16, 25, \dots\}$$

Ket : Memiliki anggota berbeda, tetapi jumlah yang sama.

Maka diketahui fungsi $f: A \rightarrow B$ bersifat satu-satu (injektif) yaitu : $f(3) = 1$, $f(\frac{3}{2}) = 4$, $f(1) = 9$, $f(\frac{3}{4}) = 16$, $f(\frac{3}{5}) = 25$. $\Rightarrow f(\frac{3}{n}) = n^2$.

$$\text{Jadi : } A \sim B$$